



CAPABILITY
PORTFOLIO
2026

REMOTE SENSING · EARTH OBSERVATION · APPLIED AI

Seeing Earth, Smarter

AI-powered Earth Observation for governments, agribusiness, climate finance and geospatial partners from national mapping agencies to insurers.

PARIS, FRANCE · BEIRUT, LEBANON

[APP.RASID.AI](https://app.rasid.ai) · [RASID.AI](https://rasid.ai)

A Boutique Geospatial & AI Consultancy

RASID turns Earth Observation data into decision-ready intelligence. We combine deep remote-sensing expertise, geospatial AI and software engineering to build production systems for governments, enterprises and international organizations.

01

11 years

11+ years of team experience across remote sensing, deep learning and time-series satellite imagery.

02

Bespoke Consultancy

We work directly with organizations to design and deliver tailored geospatial and AI solutions.

03

GoPilot

Our flagship AI geospatial platform, turning natural-language questions into executable Earth Observation workflows.

\$1.3T

GLOBAL AI MARKET BY 2030
Source: GlobalData, 2026

04

R&D Experience

Our R&D spans AI, remote sensing and GeoAI, with 60+ scientific publications and a focus on turning validated methods into operational solutions.

05

AWS GenAI Challenge Winner

GoPilot was selected as a winner of the AWS Geospatial GenAI Challenge 2026, recognizing our work at the intersection of AI and Earth Observation.

Services

RASID combines Earth Observation, GeoAI and AI engineering to deliver bespoke analyses, monitoring solutions and production systems across environmental, agricultural, urban and infrastructure applications.

01

AI Engineering & Consultancy

From AI strategy and model development to production deployment, we build and integrate AI systems across agentic AI, computer vision, RAG, OCR, data engineering and MLOps.

02

Analysis-Ready Satellite Data

Raw satellite imagery processing, quality assessment, anomaly detection, geometric and radiometric correction, orthorectification and analysis-ready data preparation.

03

Urban Monitoring

Building and settlement change, road condition, urban heat, land-use change and environmental monitoring from multi-source imagery.

04

Climate & Environmental Intelligence

Methane detection, climate-risk analysis, wildfire and flood risk, landscape monitoring and other climate-related applications.

05

Agriculture Analytics

Crop monitoring, crop mapping, wheat and banana analytics, grain-bin classification and yield forecasting.

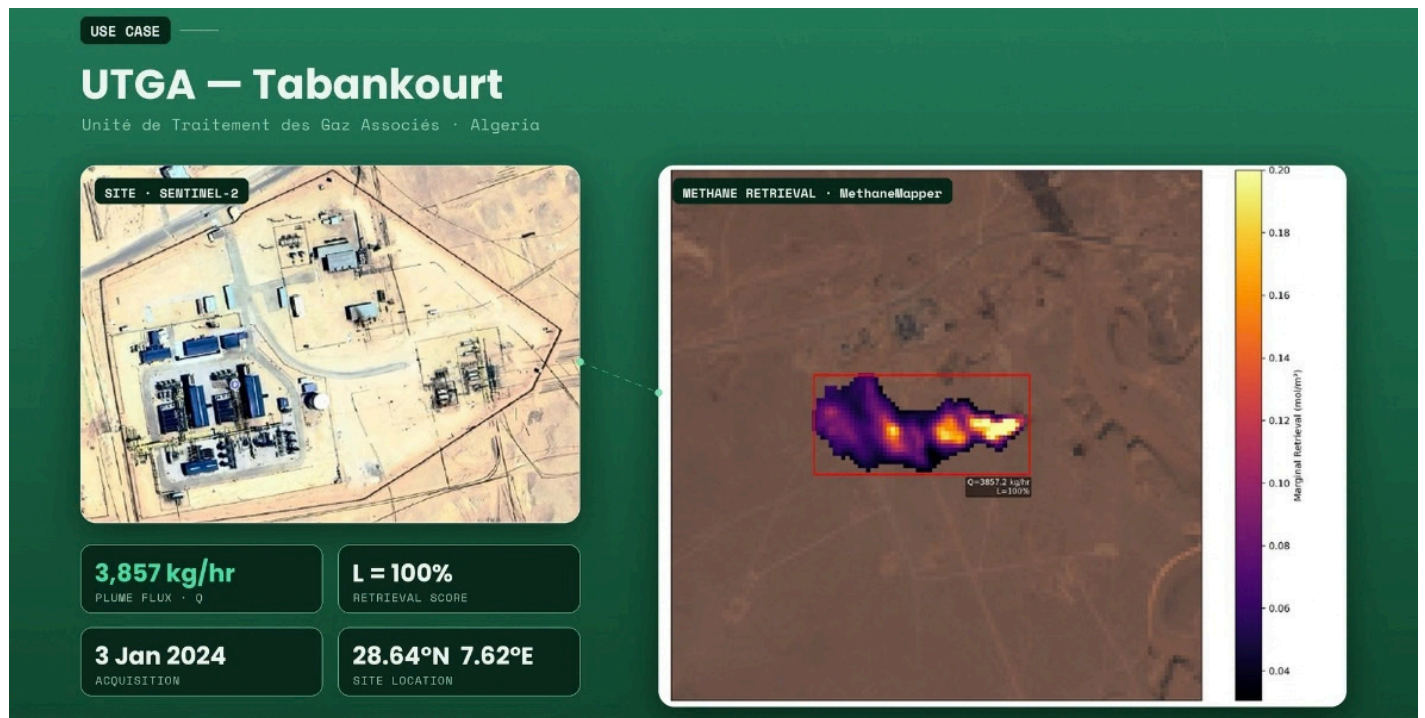
06

AI Agent Design

AI agent design integrating LLMs, tools, data and workflows to automate complex tasks.

EBRD CLIMATE CHANGE INNOVATION · Methane Monitoring

Oil&Gas Methane Plume Monitoring



OVERVIEW

AI-based methane plume detection and quantification for oil & gas operations using satellite imagery.

SOLUTION (METHODOLOGY)

Physics-informed AI detects potential methane plumes and supports rapid site-level investigation.

PROBLEM

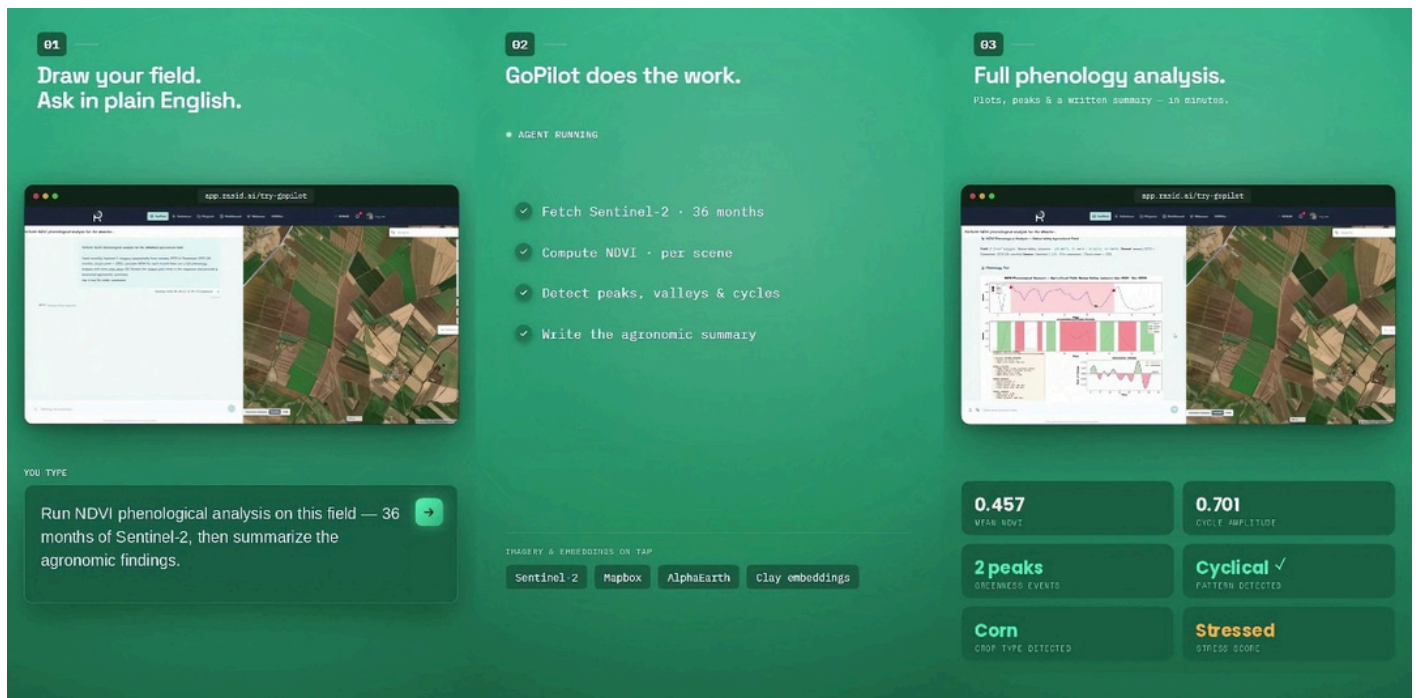
Methane leaks from oil & gas infrastructure are difficult to detect and monitor at scale.

RESULTS

80% of confirmed UNEP plume events recovered, with results delivered in minutes.

PRIVATE AGRICULTURAL CLIENT · CROP MONITORING · BANANA TR4

Banana Crop Monitoring



OVERVIEW

Satellite-based crop monitoring for banana plantations, with a focus on early detection of Fusarium Wilt (TR4).

PROBLEM

Fusarium Wilt (TR4) can devastate banana crops before visible symptoms appear, creating a need for early warning.

SOLUTION (METHODOLOGY)

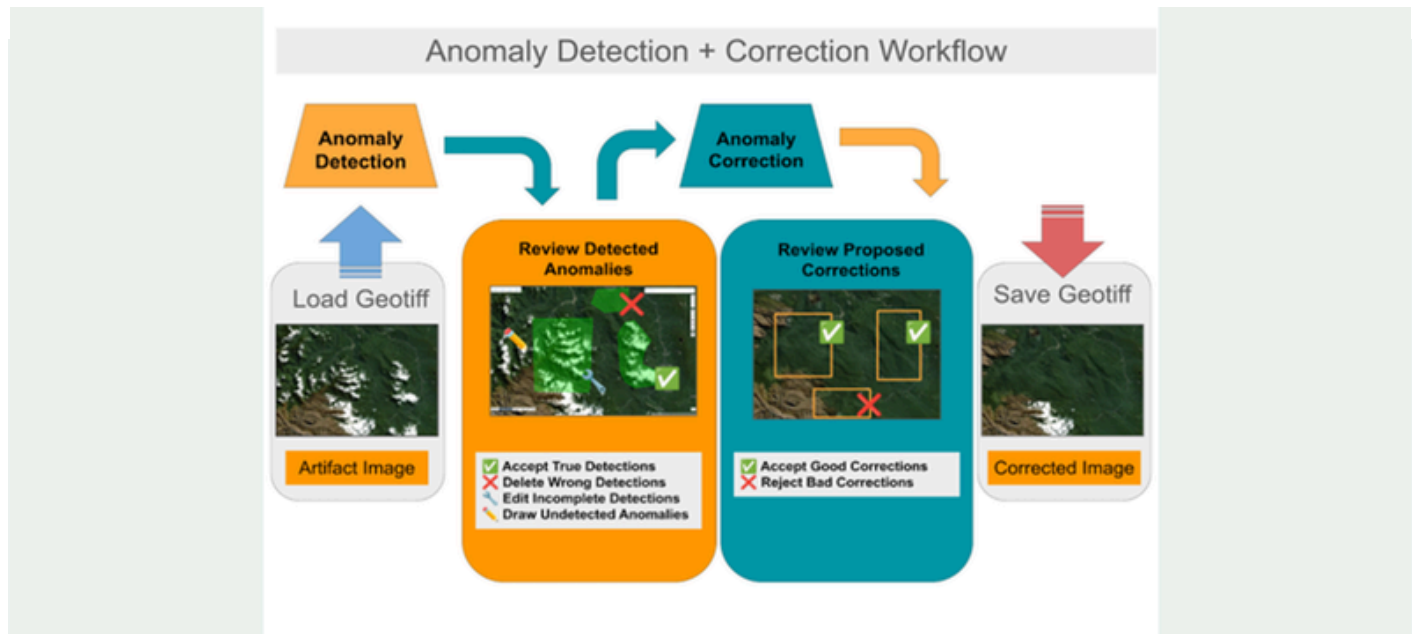
Sentinel-2 imagery is analysed to detect crop stress and potential disease signatures before visible outbreaks.

RESULTS

Operational crop-monitoring workflow combining phenology analysis with potential TR4 stress detection.

PRIVATE SATELLITE-IMAGERY PROVIDER · 2026

Analysis-Ready Satellite Data



OVERVIEW

AI-based pipeline that corrects geometric and radiometric anomalies in high-resolution satellite imagery and converts raw data into Analysis-Ready Data (ARD).

SOLUTION (METHODOLOGY)

Deep-learning co-registration, orthorectification, radiometric enhancement and automated cloud, haze and shadow detection and removal.

PROBLEM

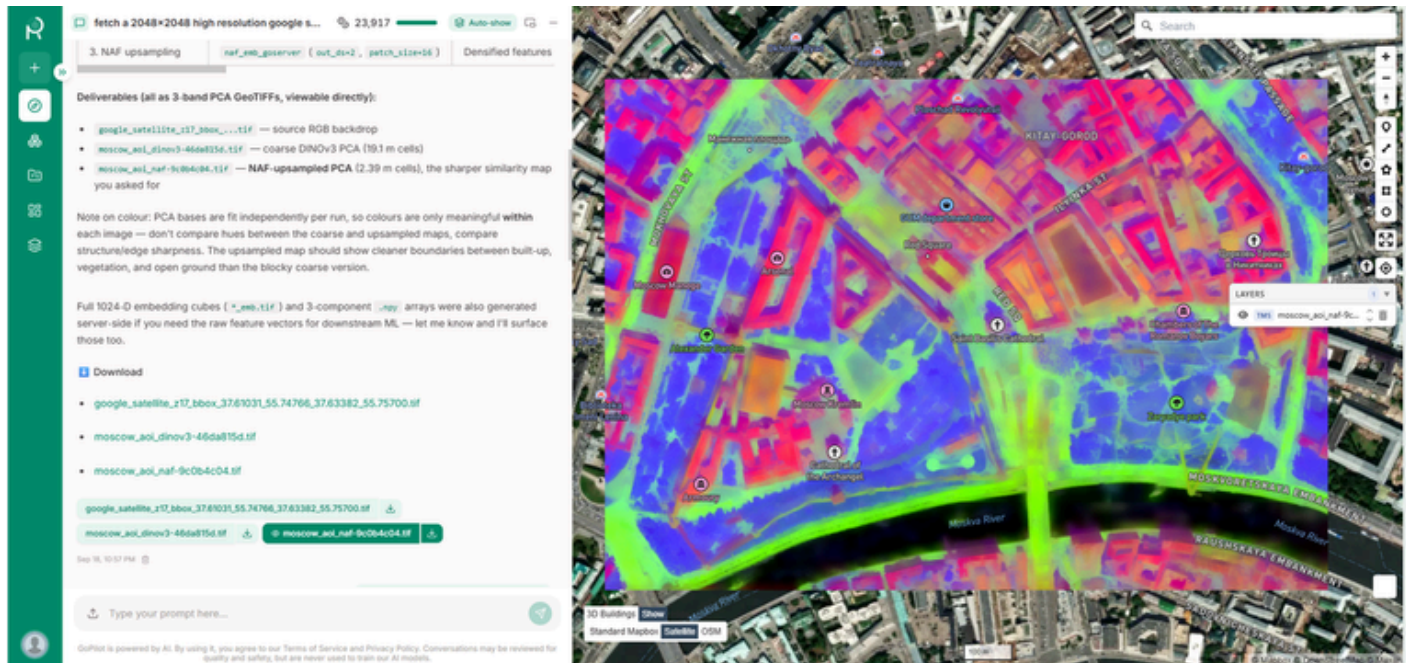
Geometric distortion, radiometric inconsistencies and atmospheric artefacts can make satellite imagery difficult to use and require costly manual correction.

RESULTS

Client-ready ARD pipeline processing ~200 GB in ~2 hours, turning raw imagery into sellable products.

AI GEOSPATIAL AGENT . AWS

GoPilot: RASID's AI Geospatial Agent



OVERVIEW

An AI geospatial agent that turns natural-language requests into executable Earth Observation workflows.

SOLUTION (METHODOLOGY)

Built on AWS Bedrock and Strands, with Claude as the LLM backbone and AI models, geospatial tools and GIS integrations connected through an agentic workflow.

PROBLEM

Geospatial analysis often requires specialist GIS skills, complex data preparation and manual scripting.

RESULTS

Award-winning AI geospatial platform, deployed through a production web application with GIS integrations.

Projects

R&D and consulting projects delivered by a team with experience across AI and geospatial technologies.

REF	PROJECT	BENEFICIARY
P08	Synthetic Methane Plume for Satellite-Based MRV	Brazil-based Client
P07	Oil & Gas Methane Plume Monitoring	EBRD Climate Change Innovation Programme
P06	Banana Crop Monitoring	Private Agricultural Client
P05	Lebanon Solar Rooftop Potential Mapping	ESA EO Data Access
P04	AI for Food Security in the Arab World	AOAD
P03	Grain Bins Classification across the USA	US-based AgriTech Client
P02	Handwritten OCR for the Public Sector	UNDP - RBAS
P01	RAG Knowledge Platform	Private Consultancy Client



Team

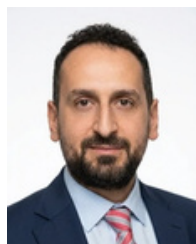
A senior team combining research, AI engineering, geospatial expertise and commercial experience directly involved in every engagement.



01

Dr. Ali J. Ghandour

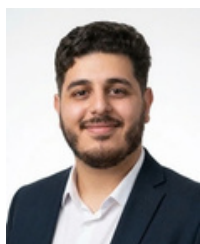
R&D AI Consultant.



02

Reda Haidar

Commercial & Strategy Director.



03

Hasan Nasrallah

AI Solutions Architect.



04

Hasan Wehbi

AI System Engineer.



05

Mohamad El Moussawi

Platform Engineer.



06

Amira El Halabi

BD & Pre-Sales.



Let's talk.

RASID helps organisations turn Earth Observation and AI into practical, production-ready solutions through bespoke consulting, custom development and our own geospatial AI platforms.

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BOOK A MEETING

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